

The equation of a straight-line:

$$y = mx + c$$

m = gradient

c = y-intercept

$$y - y_1 = m(x - x_1)$$

m = gradient

(x_1, y_1) is a point on the line

$$ax + by + c = 0$$

$$\frac{-a}{b} = \text{gradient}$$

$$\frac{-c}{b} = \text{y-intercept}$$

Properties/Uses:

Given two points (x_1, y_1) and (x_2, y_2) on a straight-line:

$$\text{Gradient} = m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

If two lines are parallel, their gradients must be equal

If two lines are perpendicular, their gradients are negative reciprocals of each other ($m_1 \times m_2 = -1$)

The perpendicular bisector for a line, intersects it at its midpoint and is perpendicular to the line

Transforming relationships:

$$y = Ax^n$$

$$\ln y = \ln(Ax^n)$$

$$\ln y = \ln A + n \ln x$$

$$\ln y = n \ln x + \ln A$$

$$Y = mX + c$$

$$Y = \ln y$$

$$m = n$$

$$X = \ln x$$

$$c = \ln A$$

$$y = Ab^x$$

$$\ln y = \ln(Ab^x)$$

$$\ln y = x \ln b + \ln A$$

$$Y = mX + c$$

$$Y = \ln y$$

$$m = \ln b$$

$$X = x$$

$$c = \ln A$$

You can also go backwards to obtain the non-linear form from linear